Denoising and SNR-Aware SpeechCueLLM for Robust Speech Emotion Recognition: Reproduction, Extension and PEFT Comparison

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Code: https://github.com/UtkarshUmap/SpeechcueLLM_Reproduce

***Abstract*—SpeechCueLLM (Findings of NAACL 2025) lets a text-only LLM recognise emotion in speech by turning acoustic measurements (volume, pitch, speaking rate and their variation) into natural-language descriptions that are added to the prompt, and fine-tuning with LoRA. We reproduce the method on IEMOCAP and MELD under a tight compute budget (two Kaggle T4 GPUs, LLaMA-2-7B with 4-bit QLoRA) and extend it in three directions: a MetricGAN+ speech-enhancement front-end, blind SNR estimation (WADA-SNR) used to write SNR-aware, hedged descriptions, and a comparison of parameter-efficient fine-tuning settings (LoRA rank, DoRA). With raw descriptions we reach a weighted F1 of 72.33 on IEMOCAP (paper: 72.6) and 69.7 on MELD (paper: 67.1). Denoising alone lowered F1 (IEMOCAP -0.7, MELD -2.4 at the last epoch); SNR-aware descriptions recovered most of the loss (MELD $+2.0$, IEMOCAP $+0.9$ over denoised-only) and ended within 0.4 F1 of raw descriptions, i.e. on par rather than better. DoRA gave the best IEMOCAP score of the project (73.73). All results are single-seed, so differences below about 0.5 F1 should not be read as real effects.**

***Index Terms*—Speech emotion recognition, large language models, speech enhancement, SNR estimation, parameter-efficient fine-tuning, reproducibility.**

I. Introduction

Speech emotion recognition (SER) identifies the emotional state conveyed by spoken language and is relevant to conversational agents, customer-service systems and educational technology. A transcript alone discards vocal cues such as pitch, loudness and speaking rate that often carry emotion independently of the words. SpeechCueLLM [1] addresses this without changing the LLM architecture: five acoustic characteristics are bucketed into categorical levels with quantile thresholds and verbalised (e.g. "very low volume with very low variation"), then inserted into the prompt together with the dialogue context. The authors report clear gains on the studio-quality IEMOCAP corpus but small or negative gains on MELD, whose audio is noisy, and list robustness to noisy audio as future work.

Problem. The conversion from audio to text silently assumes that the underlying measurements are reliable. On noisy audio the measurements are corrupted, yet the description still reads as a confident statement and may mislead the LLM. Our question is whether the pipeline can be made more robust by (a) cleaning the audio before feature extraction and (b) telling the LLM how reliable the acoustic cues are.

Contributions.

- A reproduction of SpeechCueLLM on IEMOCAP and MELD with LLaMA-2-7B on two T4 GPUs (4-bit QLoRA, 3 epochs), including the repository patches needed on a modern software stack.
- A MetricGAN+ denoising front-end and a raw-versus-denoised feature diagnostic (1000 clips per dataset) that shows which features the denoiser changes.
- Blind per-utterance SNR estimation (WADA-SNR) and SNR-aware descriptions with an explicit SNR statement and hedged wording.
- A PEFT comparison on IEMOCAP (LoRA rank sweep and DoRA) on top of the SNR-aware input.
- A data-preparation bug that produced F1 ≈ 40 on MELD, its diagnosis and its fix.

Project criterion. Code is available, so we reproduce the existing implementation and address an identified limitation (dependence on clean audio) with a proposed, experimentally evaluated solution.

Team contributions.

- Utkarsh Umap (B24CS049): reproduces the official implementation, prepares datasets and environment, verifies preprocessing and baseline results, integrates the denoising front-end, maintains the experimental pipeline.
- Kaif Raza (B24DS017): blind SNR estimation, SNR-aware description generation, extended prompt format, description-level ablations (B2 vs. B3).
- Kushal Agarwal (B24DS012): PEFT methods (LoRA rank sweep, DoRA), efficiency measurements, robustness evaluation and statistical analysis.

II. Related Work

Emotion recognition in conversation. IEMOCAP [2] and MELD [3] are the standard corpora for dyadic and multi-party emotion recognition. Audio is commonly used through self-supervised speech encoders such as wav2vec 2.0 [4] fused with text representations, which requires architectural changes. LLM-based approaches recast the task as text generation; InstructERC [5] is an example that fine-tunes an LLM on conversation text.

Acoustic cues as text. SpeechCueLLM [1] keeps the text-only LLM and verbalises five acoustic characteristics, extracted with Praat [6], as short descriptions. Its pipeline is deterministic (threshold calculation, speaker-specific thresholding, categorisation, description generation) and has no notion of measurement reliability. Our work keeps this paradigm and adds reliability information.

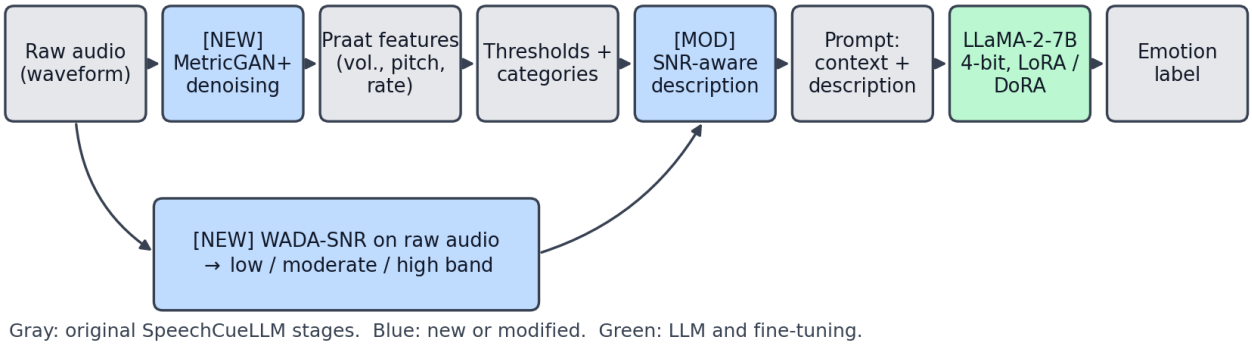


Fig. 1. Proposed pipeline. The SNR estimate is computed once per utterance on the raw audio and enters only the description stage.

Speech enhancement and SNR estimation. MetricGAN+ [7] and waveform-domain denoisers [8] remove noise, and toolkits such as SpeechBrain [9] make them easy to apply. Enhancement can also alter the signal statistics that downstream features depend on, which our diagnostic quantifies. Without clean references, SNR can be estimated blindly, for example with WADA-SNR [10].

Parameter-efficient fine-tuning. LoRA [11] trains low-rank adapters on a frozen model, QLoRA [12] does so on a 4-bit quantised backbone, and DoRA [13] decomposes weights into magnitude and direction and applies low-rank updates to the direction. We use LLaMA-2-7B [14] with 4-bit LoRA and compare ranks and DoRA.

III. Proposed Methodology

Fig. 1 shows the pipeline. Gray stages come from SpeechCueLLM; blue stages are new or modified.

A. Base pipeline

Five features are extracted per utterance (average and variation of volume, average and variation of pitch, speaking rate) with the repository's Praat-based extractor, bucketed into five levels by fixed percentile thresholds (0.1, 0.25, 0.75, 0.9), and converted into a sentence, e.g. "very low volume with very low variation, moderate pitch with very high variation, low speaking rate". The prompt contains the last utterances of the dialogue (history window 12), the description of the target utterance and the label set, and the model answers with a single label.

B. Denoising front-end (B2)

Audio is enhanced with MetricGAN+ (`speechbrain/metricgan-plus-voicebank`) before feature extraction. The whole IEMOCAP and MELD datasets were denoised on CPU and descriptions were regenerated from the denoised features. A diagnostic on a random sample of 1000 clips per dataset (seed 0) compares raw and denoised feature distributions and compares same-bucket agreement under frozen and recomputed thresholds (Section V-B).

TABLE I
SNR band edges (dB), training-set tertiles.

Dataset	Low	Moderate	High
MELD	≤ 13	≤ 19	> 19
IEMOCAP	≤ 16	≤ 23	> 23

C. Blind SNR estimation and bands

WADA-SNR [10] is computed on the raw audio of every utterance and clipped to $[-10, 40]$ dB. Because no clean reference exists for MELD, the band edges come from training-set tertiles (Table I) in the same data-driven way as the original feature thresholds.

D. SNR-aware descriptions (B3)

Relative to B2 only the description column changes. An explicit SNR statement is added and the acoustic cues are hedged with words such as "likely" or "possibly" depending on the band; context tags are left unhedged. Every feature is always included so that the prompt structure stays fixed. The SNR signal describes measurement reliability only and never refers to the emotion label. One corrupt MELD training clip (`dia125_utt3`) was assigned the lowest band; for IEMOCAP the SNR rows were joined by rebuilding utterance file names from video id, gender and per-gender running count (151 dialogue-level rows dropped).

E. Configurations and PEFT variants

B0: text only. B1: raw descriptions plus context (original method). B2: denoised descriptions plus context. B3: B2 plus SNR-aware wording. For PEFT we vary the LoRA rank ($r \in \{8, 16, 32\}$, with $\alpha = r$) and enable DoRA at $r = 16$, all on the B3 input. The paper's backbone and adapter family are retained, so no architecture change is introduced.

IV. Experimental Setup

A. Data

IEMOCAP: 10,039 utterances (Kaggle copy of the full release), six classes (happy, sad, neutral, angry, excited, frustrated). MELD: 13,848 mp4 clips converted to 16 kHz mono

TABLE II
TRAINING SETTINGS (LLAMA-2-7B).

Item	This work	Paper
Epochs	3	15
Learning rate	3e-4 IEMOCAP, 2e-4 MELD	3e-4 IEMOCAP
Effective batch	8	not matched
Max length	2500 IEMOCAP, 1500 MELD	–
History window	12	–
Seed	11	mean of 3 runs
Adapter	LoRA r16, α16, q/k/v	LoRA, 12.6M params
Precision	4-bit NF4 + fp16 compute	–

wav with ffmpeg and seven emotion classes; the corrected description files hold 9,988 / 1,112 / 2,615 rows for train / dev / test, which give 9,980 / 2,610 train / test prompts. Metric: weighted F1 on the test set.

B. Hardware, software and hyperparameters

Kaggle notebooks with two NVIDIA T4 GPUs (12 h session limit, about 30 GPU hours per week), DeepSpeed ZeRO-2 [15], fp16 (the T4 has no bf16), gradient checkpointing, Python 3.13 with recent `peft`, `deepspeed` and `transformers` instead of the repository's pinned environment. Table II lists the settings. A run takes about 4.5 h on IEMOCAP (about 1.8 s/step, 2582 steps per epoch) and about 6 h on MELD.

C. Repository patches and caveats

Deviations from the paper: T4 instead of larger GPUs, fp16 instead of bf16, 4-bit quantisation, 3 instead of 15 epochs, effective batch 8, a single seed and newer library versions. Patches to the official repository: load the model in fp16 with `low_cpu_mem_usage`; force fp16 rather than bf16; enable input gradients for gradient checkpointing; drop empty optimiser parameter groups; remove the `mics_shard_size` keys that newer DeepSpeed rejects; add `--gradient_checkpointing` to the shell script; uninstall `torchao` (clashes with `peft`); use ZeRO-2; install `opensmile`. In a committed Kaggle run one must not call `os._exit`, which marks the version as failed and loses the output.

Each of the two GPU processes scores the full test set, so two F1 values are logged per epoch that can differ by a few tenths of a point; we did not investigate the cause. The best epoch is chosen on the test set, as in the repository, which is optimistic, so we report per-epoch scores. Code, notebooks and dataset names are listed in the appendix and in the repository.

V. RESULTS AND ANALYSIS

A. Baseline reproduction

Table III compares our baselines with the paper. With descriptions and context, IEMOCAP reaches 72.33 after three epochs, 0.3 below the paper's 72.6 after fifteen. Text only reaches 71.30, 1.2 above the paper's 70.11, so our gain from descriptions on IEMOCAP is $+1.0$ against $+1.9$ in the paper. Our MELD scores are about 2 points above the paper's, and descriptions give no clear gain over text only on MELD ($+0.3$ at the best epoch, $+0.9$ at the last), in line with the paper.

TABLE III
WEIGHTED F1 PER EPOCH (OURS) AGAINST THE PAPER (LORA, 15 EPOCHS, MEAN OF 3 RUNS).

Setting	Dataset	Ours (ep. 1 / 2 / 3)	Paper
Text only (B0)	IEMOCAP	63.87 / 68.18 / 71.30	70.11
Desc. + context (B1)	IEMOCAP	64.38 / 70.23 / 72.33	72.6
Text only (B0)	MELD	69.53 / 68.14 / 68.80	67.44
Desc. + context (B1)	MELD	69.81 / 68.93 / 69.72	67.1

TABLE IV
RAW TO DENOISED FEATURES: MEDIAN SHIFT AND SPEARMAN ρ.

	MELD		IEMOCAP	
Feature	shift	ρ	shift	ρ
Volume average	+17.0%	0.346	+23.8%	0.124
Volume variation	+41.1%	0.414	+20.9%	0.626
Pitch average	+1.9%	0.810	+11.5%	0.749
Pitch variation	+3.6%	0.460	+21.5%	0.294
Speaking rate	+6.7%	0.424	+8.1%	0.455

TABLE V
SAME-BUCKET AGREEMENT (%), FROZEN / RECOMPUTED THRESHOLDS.

Feature	MELD	IEMOCAP
Volume average	11.0 / 37.4	10.6 / 35.9
Volume variation	23.8 / 44.3	28.0 / 48.2
Pitch average	67.2 / 67.5	56.7 / 55.6
Pitch variation	49.2 / 52.1	40.9 / 44.5
Speaking rate	35.4 / 42.3	40.7 / 43.4

B. Effect of denoising on acoustic features

MetricGAN+ costs about 0.058 s per clip alone and about 0.5 s per clip with ffmpeg and both feature extractions. Table IV gives the median shift and the Spearman correlation between raw and denoised features.

- The volume-average shift is enhancer gain, not a real change: after removing the per-clip RMS gain change the correlation is 0.995 (MELD) and 1.0 (IEMOCAP), and level-matching to the raw RMS gives 97.2% (MELD) and 99.2% (IEMOCAP) same-bucket agreement with frozen thresholds.
- Pitch average is the most stable feature on MELD; pitch variation and speaking rate change substantially.
- Clean IEMOCAP changes about as much as noisy MELD, so part of the change is probably denoiser artefact (for example pitch-tracker jumps; one test clip went from 21.8 to 84.2 Hz pitch standard deviation). This is a hypothesis, since there is no ground truth.

Chance same-bucket agreement for five equal classes is 31.5%. Table V shows agreement for frozen and recomputed thresholds. Frozen thresholds are close to chance for volume average and volume variation, so we recompute thresholds for volume variation, pitch variation and speaking rate, keep either option for pitch average, and handle volume average by level-matching. Thresholds are global (not speaker or gender specific) and based on a 1000-clip sample.

C. Main results: B0 to B3

Table VI and Fig. 2 give weighted F1 per epoch. At the last epoch denoising lowers F1 on both datasets relative to raw descriptions (IEMOCAP -0.7, MELD -2.4), and SNR-aware descriptions recover most of the loss (MELD $+2.0$ over B2,

TABLE VI
WEIGHTED F1 PER EPOCH, SEED 11, LLAMA-2-7B 4-BIT QLORA.

	IEMOCAP			MELD		
Config	ep.1	ep.2	ep.3	ep.1	ep.2	ep.3
B0 text only	63.87	68.18	71.30	69.53	68.14	68.80
B1 raw desc.	64.38	70.23	72.33	69.81	68.93	69.72
B2 denoised	62.29	69.05	71.60	66.16	66.22	67.34
B3 denoised + SNR	64.67	70.30	72.50	69.42	68.03	69.38

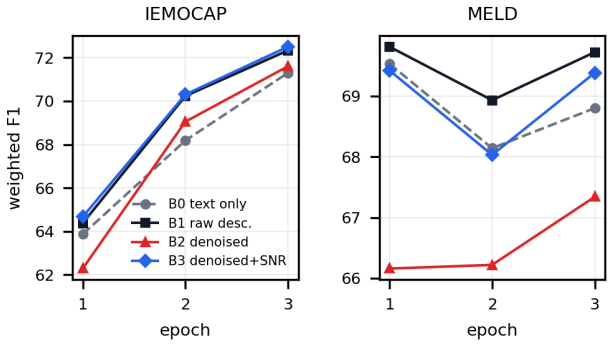


Fig. 2. Weighted F1 per epoch for B0–B3.

IEMOCAP +0.9). B3 is within 0.4 F1 of raw descriptions on both datasets, so it is on par rather than better. The IEMOCAP B3 values come from the P1 run, which uses the B3 input with the repository's default adapter (r16, α16).

D. A data bug on MELD

The first MELD B2 run scored only 40.39 / 40.14 / 39.28, far below the paper's roughly 67. The cause was in data preparation: the speaker helper files were built from float `Dialogue_ID` and `Utterance_ID` columns in the raw dev and test files, so file names did not match, every dev and test row lost its speaker and utterance text, and the context lookup in `data_process.py` broke. Prompt statistics exposed the symptom: test prompts held 4.1 utterances on average against 6.6 in train, and 1,033 test prompts had a single utterance although the test set has only about 280 dialogues. After rebuilding the files with IDs parsed from the file name (folder `B2_meld_fixed`) the test mean is 6.4 and 283 test prompts have a single utterance. The IEMOCAP B2 run was never affected.

E. PEFT comparison (IEMOCAP)

Setup: B3 input, seed 11, 3 epochs, learning rate 3e-4, ZeRO-2, fp16, gradient checkpointing, q/k/v LoRA with $\alpha = r$; P1 is the repository default r16. The comparison is restricted to IEMOCAP because of compute limits.

Rank has little effect in this range: r8 is within 0.5 F1 of r16 (inside the run-to-run spread of about 0.4) with half the parameters. DoRA is best at every epoch (+1.2 F1 over r16 at epoch 3) for about 26% more training time. GPU memory stayed under 7 GB per GPU in all runs, so memory does not separate the methods. The r32 result is still to be added (TBD); r4 was proposed but not run.

TABLE VII
PEFT RUNS. PARAMETER COUNTS ARE COMPUTED FROM THE LORA CONFIGURATION, NOT READ FROM THE LOGS.

Run	Method (params)	ep.1	ep.2	ep.3	Time (s)
P1	LoRA r16 (12.6M)	64.67	70.30	72.50	16,234
P2	LoRA r8 (6.3M)	63.35	70.75	72.98	16,529
P3	LoRA r32 (25.2M)	TBD	TBD	TBD	TBD
P4	DoRA r16 (13.0M)	65.03	71.33	73.73	20,436

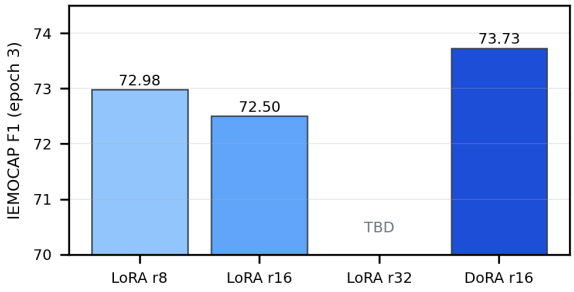


Fig. 3. IEMOCAP weighted F1 at epoch 3 for the PEFT settings (truncated y-axis).

TABLE VIII
CHANGE IN WEIGHTED F1 (POINTS) FROM EACH COMPONENT.

Component (comparison)	IEMOCAP last = best	MELD last	MELD best
Descriptions (B1−B0)	+1.03	+0.92	+0.28
Denoising (B2−B1)	−0.73	−2.38	−2.47
SNR-aware wording (B3−B2)	+0.90	+2.04	+2.08
Net effect (B3−B1)	+0.17	−0.34	−0.39

VI. ABLATION STUDY

Table VIII isolates the effect of each component by differencing configurations that differ in one factor: B1−B0 isolates descriptions, B2−B1 isolates denoising, B3−B2 isolates SNR-aware wording.

The PEFT ablation gives r8 − r16 = +0.48 and DoRA − LoRA (r16) = +1.23 at epoch 3. Two ablations were not run: separating the SNR statement from the hedge words, and training with level-matched denoised features and recomputed thresholds. With one seed per cell, the IEMOCAP differences below about 0.5 and the MELD net effect (B3−B1) are not distinguishable from noise.

VII. DISCUSSION

Reproduction. The method reproduces: IEMOCAP with raw descriptions reached 72.33 after three epochs against the paper's 72.6 after fifteen. The MELD gap in our favour (about 2 points) is probably due to training schedule, quantisation and library versions, not a real improvement.

Descriptions help little on MELD. This agrees with the paper. On IEMOCAP the gain from descriptions is +1.0 in our setting.

Denoising is not free. MetricGAN+ shifts pitch variation and speaking rate materially (Table IV) and lowered downstream F1, most on noisy MELD. Part of the volume shift is plain gain and can be removed by level-matching.

Telling the model about audio quality helps. SNR-aware hedging recovered the denoising loss, especially on MELD. Our interpretation, which we did not test, is that the model learns to discount unreliable cues when the prompt says they are unreliable.

PEFT. Adapter size matters little between r8 and r16; DoRA gives a small gain for extra compute and is a candidate for the final configuration.

VIII. Limitations

- Single seed per condition (the paper averages three); differences below about 0.5 F1 should not be read as real effects.
- The best epoch is selected on the test set (optimistic), and the curves were still rising at epoch 3, so three epochs under-train relative to the paper's fifteen.
- Global rather than speaker-specific thresholds, and a 1000-clip sample for the threshold analysis.
- The PEFT study covers IEMOCAP only, the learning rate was not retuned per rank, DoRA was not run on MELD, and the r32 run is missing.
- There is no ground truth to separate denoiser artefacts from true feature changes, and no ablation separates the SNR statement from the hedge words.
- The robustness evaluation across artificial noise levels proposed in the project plan was not carried out.
- The IEMOCAP B3 scores are taken from the P1 run rather than a separately logged run.

IX. Conclusion and Future Work

We reproduced SpeechCueLLM with LLaMA-2-7B on modest hardware, showed that a MetricGAN+ front-end alters pitch variation and speaking rate enough to lower downstream F1, and found that SNR-aware descriptions recover most of that loss, ending on par with raw descriptions. DoRA gave the best IEMOCAP score (73.73) at about 26% more training time. The negative result for denoising alone is reported as found.

Future work: second and third seeds with mean and standard deviation; a robustness study with artificial noise at several SNR levels (ΔF1 and relative robustness); an ablation of the SNR statement versus hedge words; training with level-matched denoised features and recomputed thresholds; speaker- and gender-specific thresholds; PEFT on MELD and the missing r4 and r32 points; investigating why the two GPU processes disagree; and longer training up to the paper's 15 epochs.

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Appendix

Appendix A: AI Tools and Prompts

AI assistants (ChatGPT and Claude) were used to understand background concepts (speech enhancement, blind SNR estimation, PEFT methods, the SpeechCueLLM pipeline), to debug the Kaggle notebooks, and to organise and edit this report; all experiments were run and checked by the team. Platforms: Kaggle (training), GitHub (version control), Hugging Face (model weights), VS Code. Representative prompts:

- "Explain the SpeechCueLLM paper step by step in simple terms."
- "What is the final output of the SpeechCueLLM model, and how is it evaluated?"
- "What role does LoRA play in this paper, and how does it differ from full fine-tuning?"
- "What are practical, lightweight speech-enhancement models suitable for a 16–24 GB GPU?"
- "What are standard blind SNR estimation methods for real-world noisy audio?"
- "Compare QLoRA, LoRA rank variation, and DoRA as parameter-efficient fine-tuning strategies."
- "How can an SNR estimate be used to modulate confidence in categorical feature descriptions fed to an LLM?"
- "Should acoustic-feature thresholds be recomputed on denoised audio or kept from the raw-audio computation?"
- "Guide me in sequence through running the Kaggle notebook my teammate sent for the IEMOCAP B2 run."

- “Why does training finish but crash at evaluation with a CUDA device-side assert?”
- “Why is MELD F1 about 40 when the paper reports about 67? Which free checks can find the cause?”
- “After B0–B3 are done, what remains in the project plan and in what order?”

APPENDIX B: REPRODUCIBILITY

- Code and all notebooks: https://github.com/UtkarshUmap/SpeechcueLLM_Reproduce. Baseline code: https://github.com/zehuiwu/SpeechCueLLM [16].
- Data (Kaggle): `speechcuellm-denoise-diag`; `kushalag21/iemocap-denoised-dataset` (`B2/`, `B3_iemocap/`); `kushalag21/meld-denoised-fixed` (`B2_meld_fixed/`); `kushalag21/meld-b3` (`B3_meld/`); WADA-SNR files `iemocap_wada_snr.csv` and `meld_wada_snr.csv`.
- Models: LLaMA-2-7B-hf (`meta-llama/Llama-2-7b-hf`), MetricGAN+ (`speechbrain/metricgan-plus-voicebank`).
- Seed 11 for all reported runs; hyperparameters in Table II.